

Disorder in Cavity-Coupled Jahn–Teller Active Molecules

From Single-Molecule Polaritons to Disorder-Averaged Collective Spectra

Summer Internship Report

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Declaration

I hereby declare that I have completed an internship under the guidance of Prof. Krishna R. Nandipati (Department of Chemistry) and Prof. Athreya Shankar (Department of Physics), Indian Institute of Technology Madras. I am submitting a brief overview of my project to the National Institute of Science Education and Research (NISER), Bhubaneswar.

Signature of the Student
Date:

The internship work reported in this document entitled **Disorder in Cavity-Coupled Jahn–Teller Active Molecules** was carried out under our supervision.

Signature of the Supervisor (Prof. Krishna R. Nandipati)
Date:

Signature of the Supervisor (Prof. Athreya Shankar)
Date:

1 Cavity–matter interaction

A Fabry–Pérot (FP) cavity is a pair of parallel mirrors separated by a distance L (Fig. 1). The mirrors quantise the electromagnetic field inside into a discrete set of standing-wave modes of frequency ω_c fixed by L , concentrating the vacuum field into a small mode volume, which strongly enhances its interaction with any matter placed inside.

When the light–matter coupling exceeds every loss rate in the problem (mirror transmission, absorption, molecular linewidth), the system enters the *strong coupling* regime [11]: molecular excitation and cavity photon can no longer be treated separately but hybridise into **polaritons**, coherent superpositions of matter and light. The two hybrid states - the *lower* (LP) and *upper* (UP) polariton - are split by the **Rabi splitting** Ω . Because they carry both molecular and photonic character, strong coupling has become a route to modifying photophysics and photochemistry without chemically altering the molecules [11, 12, 13].

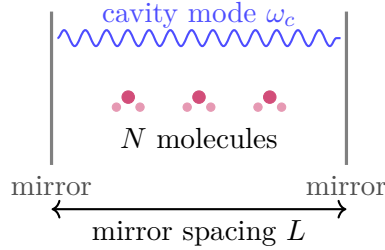


Figure 1: N molecules inside a Fabry–Pérot cavity. The mirrors quantise the field into discrete standing-wave modes of frequency ω_c ; in the strong-coupling regime molecule and photon hybridise into polaritons.

The coupling in molecular polaritonics is also *collective*: for N identical molecules the bright (symmetric) superposition of molecular excitations couples to the photon with a strength enhanced by $\sqrt{N}$, so the Rabi splitting scales as $\Omega = 2\sqrt{N}g$ with $g = \Omega/(2\sqrt{N})$ the *per-molecule* coupling. The remaining $N - 1$ orthogonal combinations carry no photon weight at all: they are the **dark states**. This bright/dark separation is exact only for *identical* molecules, and its breakdown under disorder is the subject of Secs. 5–7.

2 Jahn–Teller active molecules and the $(E \times e)$ model

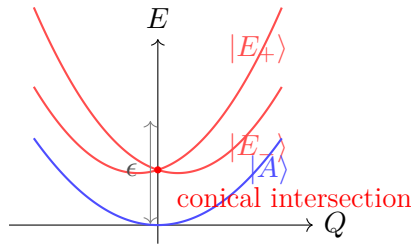


Figure 2: Cut through the $(E \times e)$ potential energy surfaces along one component of the degenerate mode coordinate Q : ground state $|A\rangle$ (blue) and the JT-split excited doublet (red), which touch at a conical intersection. In the full two-dimensional (Q_x, Q_y) space the lower excited sheet is the “Mexican hat”.

The **Jahn–Teller (JT) theorem** [5] states that a non-linear molecule in an orbitally degenerate electronic state is unstable: it spontaneously distorts along a non-totally-symmetric coordinate so as to lift the degeneracy. A standard example is the $(E \times e)$ **problem** [6, 7, 8], in which a doubly degenerate electronic state E (components $|E_x\rangle, |E_y\rangle$) couples to a doubly degenerate vibrational mode e with coordinates (Q_x, Q_y) .

Molecules with an n -fold rotational axis $C_{n \geq 3}$ possess such doubly degenerate irreducible representations [1]; planar aromatics such as sym-triazine (D_{3h}) and benzene (D_{6h}) are stan-

standard examples [9, 10], and the sym-triazine parameter set is used throughout this work. Two properties of the E subspace matter here: by selection rules the two orthogonal polarisation directions of light propagating along the C_n axis couple the totally symmetric ground state $|A\rangle$ to the corresponding $E_{x/y}$ component, and the vibronic coupling between $|E_x\rangle$ and $|E_y\rangle$ mixes their electronic character.

It is convenient to work in the complex electronic basis

$$|E_{\pm}\rangle = \frac{|E_x\rangle \pm i|E_y\rangle}{\sqrt{2}}, \quad (1)$$

in which the JT coupling has a transparent angular-momentum interpretation. The adiabatic potential energy surfaces of the excited doublet form a *Mexican hat* with a **conical intersection** at the reference geometry Q_0 (Fig. 2); the nuclear motion around the brim is a *pseudorotation* that carries vibrational angular momentum, which is what allows JT systems to exchange angular momentum with circularly polarised light [16].

3 Single molecule: cavity Jahn–Teller polaritons

This section reproduces the model and the main spectroscopic result of Nandipati and Vendrell [1].

3.1 The cavity–Jahn–Teller Hamiltonian

The molecule is placed inside the FP cavity with its C_n axis along the cavity axis z , so that the two normal-incidence cavity modes with polarisations $(\mathbf{e}_x, \mathbf{e}_y)$ couple to the E_x and E_y excitations respectively. The cavity-JT (CJT) Hamiltonian is

$$\hat{H} = \hat{H}_{\text{JT}} + \hat{H}_C + \hat{H}_{CM}, \quad (2)$$

where $\hat{H}_{\text{JT}}$ is the linear $(E \times e)$ JT Hamiltonian, $\hat{H}_C$ the cavity Hamiltonian, and $\hat{H}_{CM}$ the light–matter coupling. In the electronic basis $\{|E_+\rangle, |A\rangle, |E_-\rangle\}$ the molecular JT Hamiltonian reads [1]

$$\hat{H}_{\text{JT}} = \sum_{r=+,-} \left[\hbar\omega \hat{b}_r^\dagger \hat{b}_r + \epsilon (|E_r\rangle \langle E_r|) \right] + \frac{\kappa}{\sqrt{2}} \left\{ (\hat{b}_+^\dagger + \hat{b}_-) |E_-\rangle \langle E_+| + \text{H.c.} \right\}, \quad (3)$$

where ω , ϵ and κ are respectively the frequency of the e modes, the energy of the E electronic states at the reference geometry Q_0 , and the linear JT coupling parameter. The vibrational ladder operators are the circular combinations

$$\hat{b}_{\pm} = \frac{\hat{b}_x \mp i\hat{b}_y}{\sqrt{2}}, \quad \hat{b}_{\pm}^\dagger = \frac{\hat{b}_x^\dagger \pm i\hat{b}_y^\dagger}{\sqrt{2}}, \quad (4)$$

with $\hat{b}_x, \hat{b}_y$ the ladder operators of the real degenerate modes Q_x, Q_y .

Written in this form, Eq. (3) makes explicit that the vibronic coupling *exchanges angular momentum* between the electronic subspace and the vibrational pseudorotation. The vibrational angular momentum perpendicular to the (x, y) plane, and the electronic angular-momentum-like operator inside the E subspace, are

$$\hat{L}_z = \hbar(\hat{b}_+^\dagger \hat{b}_+ - \hat{b}_-^\dagger \hat{b}_-), \quad \hat{S}_z = \hbar(|E_+\rangle \langle E_+| - |E_-\rangle \langle E_-|), \quad (5)$$

the latter with eigenvalues $0, \pm\hbar$. The *vibronic* angular momentum of the molecular JT subsystem is then $\hat{J}_{\text{JT}} = 2\hat{L}_z + \hat{S}_z$, the factor 2 following from the π -radian (rather than 2π)

periodicity of the pseudorotation [6]. One checks that $[\hat{H}_{\text{JT}}, \hat{J}_{\text{JT}}] = 0$ - the symmetry of the linear $(E \times e)$ Hamiltonian responsible for the double degeneracy of the vibronic spectrum.

The cavity and the light-matter coupling are treated within the Condon approximation of constant transition dipole and the rotating-wave approximation:

$$\begin{aligned}\hat{H}_C &= \hbar\omega_c(\hat{a}_+^\dagger\hat{a}_+ + \hat{a}_-^\dagger\hat{a}_-), \\ \hat{H}_{CM} &= \frac{\Omega}{2}(\hat{a}_+^\dagger|A\rangle\langle E_+| + \hat{a}_-^\dagger|A\rangle\langle E_-| + \text{H.c.}),\end{aligned}\quad (6)$$

where $\hbar\omega_c$ is the cavity photon energy and $\Omega/2$ the molecule-cavity coupling in energy units, so that at zero detuning the Rabi splitting equals Ω . As for the vibrations, circular cavity modes are built from the linear polarisations, $\hat{a}_\pm = (\hat{a}_x \mp i\hat{a}_y)/\sqrt{2}$ and $\hat{a}_\pm^\dagger = (\hat{a}_x^\dagger \pm i\hat{a}_y^\dagger)/\sqrt{2}$ [15]; these carry photonic angular momentum $\pm\hbar$ and are referred to as the cavity *polarizations*. The standing circular modes of a conventional-mirror cavity have a well-defined angular momentum but no net helicity, since the helicities of the two counter-propagating photon modes forming the standing wave cancel [14]. The z component of the photonic angular momentum is

$$\hat{l}_z = \hbar(\hat{a}_+^\dagger\hat{a}_+ - \hat{a}_-^\dagger\hat{a}_-). \quad (7)$$

Conservation law. Defining the total vibronic-photonic angular momentum

$$\boxed{\hat{J} = 2\hat{L}_z + \hat{S}_z + \hat{l}_z}, \quad (8)$$

one finds $[\hat{H}, \hat{J}] = 0$ [1]. The eigenstates of the CJT Hamiltonian - the **JT polaritons** - are therefore labelled by $j = 2L + s + l$, with $L = n_+ - n_-$ the pseudorotational quantum number ($n_\pm = 0, 1, \dots$ circular vibrational quanta), $s = 0, \pm 1$ labelling the $|A\rangle$ and $|E_\pm\rangle$ electronic states, and $l = p_+ - p_-$ where $p_\pm = 0, 1$ are the circularly polarised photon occupations. States are doubly degenerate with $j = \pm 1$; absorbing a right-circularly polarised (RCP) photon from the overall ground state reaches the $j = -1$ block, studied throughout.

3.2 Result: the single-molecule polaritonic spectrum

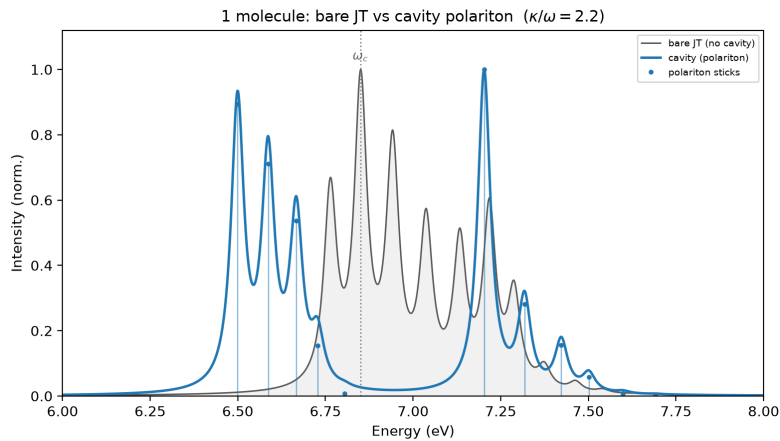


Figure 3: Reproduction of the single-molecule cavity-JT polariton spectrum [1]. Grey: the bare $(E \times e)$ JT absorption spectrum of sym-triazine (no cavity). Blue: the cavity (polaritonic) spectrum, with the polaritonic stick spectrum underneath. Parameters $\epsilon = 7$ eV, $\omega = 0.082$ eV, $\kappa/\omega = 2.2$, $\omega_c = 6.85$ eV, $\Omega/2\omega_c = 0.05$. The cavity is tuned to the brightest bare JT line; the spectrum splits into LP and UP branches.

The absorption spectrum follows from diagonalising $\hat{H}$ in the $(n_{\text{ex}} = 1, j = -1)$ block and taking the intensity of each eigenstate as its overlap with the initial state (molecule in its

vibronic ground state plus one RCP photon). Figure 3 shows the result for the sym-triazine parameters $\epsilon = 7$ eV, $\omega = 0.082$ eV, $\kappa/\omega = 2.2$ [9], with $\omega_c = 6.85$ eV resonant with the brightest bare JT transition and $\Omega/2\omega_c = 0.05$. Switching the cavity on redistributes the bare JT oscillator strength (grey) into two well-separated **LP** and **UP** branches of mixed vibronic–photonic peaks. What matters for what follows is that at $N = 1$ conservation of j together with $n_{\text{ex}} = 1$ restricts the molecule to only **three vibronic sectors**, $v \in \{0, -1, -2\}$: the range is *bounded*.

4 Collective vibronic cascade for N molecules

This section reproduces the model and the key observables of Pandit, Pandey, Shankar and Nandipati [2].

4.1 Model

We now consider N JT active molecules coupled identically to the two orthogonally polarised modes of the FP cavity. Each molecule k has two degenerate excited electronic levels well separated by an energy $\epsilon = \hbar\omega_\epsilon$ from the ground state $|A\rangle$. In terms of the linear combinations $|E_\pm\rangle = (|E_x\rangle \pm i|E_y\rangle)/\sqrt{2}$ and $\hat{b}_\pm^{(\dagger)} = (\hat{b}_x \pm i\hat{b}_y)^{(\dagger)}/\sqrt{2}$, the Hamiltonian for molecule k is ($\hbar = 1$)

$$\hat{H}_{m,k} = \sum_{r=+,-} \left[\omega \hat{b}_{r,k}^\dagger \hat{b}_{r,k} + \omega_\epsilon |E_r\rangle_k \langle E_r|_k \right] + \kappa \left[(\hat{b}_{+,k}^\dagger + \hat{b}_{-,k}) |E_-\rangle_k \langle E_+|_k + \text{h.c.} \right], \quad (9)$$

the second line being the linear ($E \times e$) JT interaction of strength κ . Coupling the $|A\rangle \leftrightarrow |E_\pm\rangle$ transitions to the two circularly polarised cavity modes within the rotating-wave approximation, the total Hamiltonian is

$$\hat{H} = \hat{H}_c + \sum_{k=1}^N (\hat{H}_{m,k} + \hat{H}_{m-c,k}), \quad (10)$$

with the free cavity Hamiltonian $\hat{H}_c = \omega_c(\hat{a}_+^\dagger \hat{a}_+ + \hat{a}_-^\dagger \hat{a}_-)$ and

$$\hat{H}_{m-c,k} = \frac{\Omega}{2\sqrt{N}} (\hat{a}_+^\dagger |A\rangle_k \langle E_+|_k + \hat{a}_-^\dagger |A\rangle_k \langle E_-|_k + \text{h.c.}). \quad (11)$$

The normalisation by $\sqrt{N}$ compensates the collective enhancement of the molecule–cavity coupling, so that the per-molecule strength $\Omega/(2\sqrt{N}) = g$ is approximately independent of N ; note that Eq. (11) reduces to Eq. (6) at $N = 1$.

4.2 Vibronic angular momentum, conserved quantities and sectors

Defining the electronic and vibrational angular-momentum operators $\hat{S}_{z,k} = |E_+\rangle_k \langle E_+|_k - |E_-\rangle_k \langle E_-|_k$ and $\hat{L}_{z,k} = \hat{b}_+^\dagger \hat{b}_+ - \hat{b}_-^\dagger \hat{b}_-$, the molecular Hamiltonian commutes with the total *vibronic* angular momentum operator

$$\hat{V}_k = 2\hat{L}_{z,k} + \hat{S}_{z,k}. \quad (12)$$

Its eigenstates therefore carry a well-defined vibronic quantum number v and group into *sectors* of fixed v , in terms of which

$$\hat{H}_{m,k} = \sum_v \sum_i \lambda_i^v |\lambda_i^v\rangle_k \langle \lambda_i^v|_k, \quad (13)$$

where λ_i^v and $|\lambda_i^v\rangle$ are the i th eigenfrequency and eigenstate in sector v . The ground manifold consists of trivial eigenstates $|A, n_+, n_-\rangle$, populating only even v since $\hat{V}_k |A, n_+, n_-\rangle = 2(n_+ - n_-) |A, n_+, n_-\rangle$; the excited manifold consists of non-trivial JT eigenstates which, because $\hat{S}_z |E_\pm\rangle = (\pm 1) |E_\pm\rangle$, populate only odd v .

The total Hamiltonian (10) conserves two quantum numbers: the total number of electronic and photonic excitations,

$$\hat{N}_{\text{ex}} = \sum_{k=1}^N (|E_+\rangle_k \langle E_+|_k + |E_-\rangle_k \langle E_-|_k) + \hat{a}_+^\dagger \hat{a}_+ + \hat{a}_-^\dagger \hat{a}_-. \quad (14)$$

Second, introducing the photonic angular momentum $\hat{P} = \hat{a}_+^\dagger \hat{a}_+ - \hat{a}_-^\dagger \hat{a}_-$, the total angular momentum

$$\hat{J} = \sum_{k=1}^N \hat{V}_k + \hat{P}. \quad (15)$$

These follow respectively from the Jaynes–Cummings nature of the cavity–molecule interaction and from the dipole selection rules, and let the Hamiltonian be block-diagonalised into sectors of fixed (n_{ex}, j) . All results below are obtained in the optically relevant single-excitation sector ($n_{\text{ex}} = 1$, $j = -1$).

4.3 The collective vibronic cascade

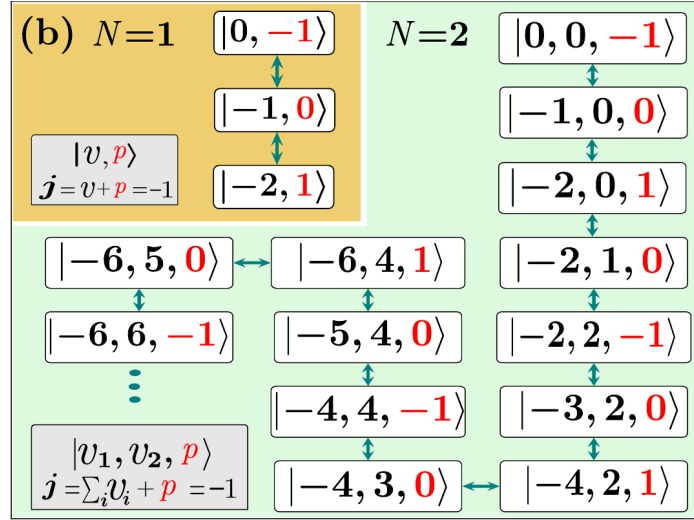


Figure 4: Reproduction of Fig. 2(b) of Ref. [2]: basis states populated starting from all molecules in the vibronic ground state plus one RCP photon, in the notation $|v, p\rangle$ ($N = 1$) and $|v_1, v_2, p\rangle$ ($N = 2$). For $N = 1$ the chain terminates after three sectors; for $N \geq 2$ there is no restriction on the individual v_k and the ladder is unbounded - the collective vibronic cascade.

For a *single* molecule in the $(1, -1)$ sector the photon carries $p = -1$ (RCP) or $p = 0$, and conservation of $j = -1$ forces $v \in \{0, -1, -2\}$ - the bounded range behind the modest structure of Fig. 3.

For *two or more* molecules this changes qualitatively. The conserved quantity is the *total* vibronic momentum $v = \sum_k v_k$, still restricted to $\{0, -1, -2\}$, but the individual v_k are now unconstrained: one molecule can be driven to large positive v_1 provided the other compensates with large negative v_2 . Since the cavity photon is repeatedly absorbed and re-emitted by different molecules, it *mediates* the transfer of angular momentum between them, generating the unbounded ladder of Fig. 4: the **collective vibronic cascade** [2].

4.4 Quantifying the cascade: sector population $P(v)$ and participation ratio

To quantify how far the cascade spreads, each polaritonic eigenstate is decomposed over the single-molecule vibronic sectors. For $N = 1$ these eigenstates have the form

$$|\psi\rangle = \sum_{v,i,p} C_{i,p}^v |\lambda_i^v, p\rangle, \quad (16)$$

where $p = 0, \pm 1$ indicates no cavity photon or one photon in the LCP/RCP mode, so that the occupation of sector v_0 is simply

$$P(v) = \sum_{i,p} |C_{i,p}^{v_0}|^2. \quad (17)$$

For $N = 2$ the permutation-symmetric eigenstates take the form

$$\begin{aligned} |\psi\rangle = & \sum_{v,i,p} C_{i,p}^v |\lambda_i^v, \lambda_i^v, p\rangle + \sum_{\substack{v,i,j,p \\ j>i}} C_{i,j,p}^v \frac{1}{\sqrt{2}} (|\lambda_i^v, \lambda_j^v\rangle + |\lambda_j^v, \lambda_i^v\rangle) |p\rangle \\ & + \sum_{\substack{v_1,v_2,i,j,p \\ v_2>v_1}} C_{i,j,p}^{v_1,v_2} \frac{1}{\sqrt{2}} (|\lambda_i^{v_1}, \lambda_j^{v_2}\rangle + |\lambda_j^{v_2}, \lambda_i^{v_1}\rangle) |p\rangle, \end{aligned} \quad (18)$$

and the probability of finding, say, molecule 1 in sector v_0 is

$$P(v_0) = \sum_{i,p} |C_{i,p}^{v_0}|^2 + \sum_{\substack{i,j,p \\ j>i}} |C_{i,j,p}^{v_0}|^2 + \frac{1}{2} \sum_{\substack{v,i,j,p \\ v<v_0}} |C_{i,j,p}^{v,v_0}|^2 + \frac{1}{2} \sum_{\substack{v,i,j,p \\ v>v_0}} |C_{i,j,p}^{v_0,v}|^2. \quad (19)$$

The factor $1/2$ accounts for the symmetric structure of the two-molecule state; by construction $\sum_v P(v) = 1$. The spread over sectors is measured by the **participation ratio**

$$\text{PR} = \frac{1}{\sum_v P^2(v)}. \quad (20)$$

$\text{PR} = 1$ means the state lives in a single sector; larger values mean it is spread over many. For $N = 1$ the bound $v \in \{0, -1, -2\}$ implies $\text{PR} \leq 3$, whereas for $N = 2$ values up to $\text{PR} \approx 20$ occur. The $P(v)$ **heatmap** plots, for every *bright* polaritonic eigenstate (x axis, energy ordered), the distribution $P(v)$ over sectors v (y axis).

4.5 Results: two-molecule spectra and the clean $P(v)$ heatmap

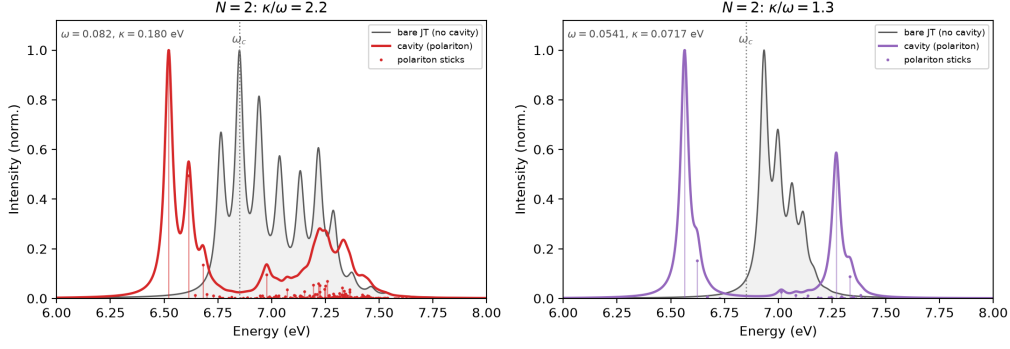


Figure 5: Reproduction of the two-molecule cavity-JT polaritonic spectra [2]. Grey: bare JT spectrum (no cavity). Coloured: $N = 2$ cavity polariton spectrum with the polaritonic stick spectrum (dots mark eigenstates with intensity above a 10^{-2} relative tolerance). **Left:** $\kappa/\omega = 2.2$ ($\omega = 0.082$ eV, $\kappa = 0.180$ eV). **Right:** $\kappa/\omega = 1.3$ ($\omega = 0.0541$ eV, $\kappa = 0.0717$ eV). Both at $\omega_c = 6.85$ eV and $\Omega/2\omega_c = 0.05$. The UP branch is visibly broadened relative to the single-molecule case of Fig. 3.



Figure 6: Clean ($\sigma = 0$) $N = 2$ vibronic-sector heatmap, computed at $\kappa/\omega = 2.2$: the single-molecule occupation probability $P(v)$ of each bright polaritonic eigenstate. x : bright states in order of increasing energy; y : sector v ; colour: $P(v)$. Vibrational cutoff $N_v = 18$. There are 42 bright eigenstates, $\text{PR}_{\text{max}} = 19.9$, $\text{PR}_{\text{mean}} = 12.7$, and sectors $v \in [-14, 13]$ are populated. This is the reference against which every disordered case in Sec. 7 is compared.

Figure 5 shows the $N = 2$ polaritonic spectra for two vibronic coupling strengths. Compared with $N = 1$ (Fig. 3), the LP branch remains fairly well resolved but the UP branch broadens into many short stems - a clear signature of the cascade, since a large number of polaritonic states form that respond only weakly to cavity driving. Figure 6 makes the mechanism explicit: the bright states spread over $v \in [-14, 13]$ with PR reaching ≈ 20 , versus the hard bound $\text{PR} \leq 3$ of the single molecule. This heatmap is the quantitative baseline for the whole disorder study.

5 Introducing disorder

5.1 Why disorder?

Both models above assume N *identical* molecules. Real ensembles are inhomogeneous: local environments, conformations and strain shift the electronic transition energy from molecule to molecule. This static *energy disorder* is the standard explanation for inhomogeneous linewidths, and it destroys the permutation symmetry underlying the clean bright/dark separation of Sec. 1. The outlook of Ref. [2] names the inclusion of disorder in molecular energies as the natural next step for the cavity-JT problem; that is the gap this work addresses.

5.2 The disordered Holstein–Tavis–Cummings model

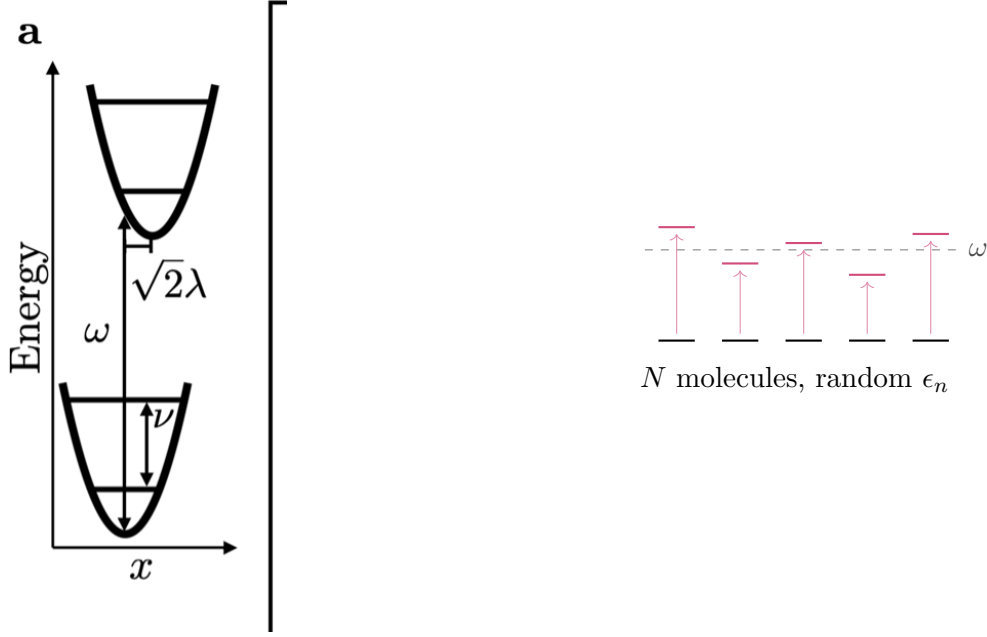


Figure 7: **Left:** a two-level molecule on displaced harmonic surfaces (vibrational spacing ν , displacement $\sqrt{2}\lambda$) - the Holstein model. **Right:** an ensemble of such molecules with random transition energies $\omega + \epsilon_n$, the static energy disorder of Ref. [4].

The clearest framework for what disorder does to a cavity-coupled molecular ensemble *with vibrations* is the **disordered Holstein–Tavis–Cummings (HTC)** model of Wellnitz, Pupillo and Schachenmayer [4]: each molecule is a two-level system on a displaced harmonic potential (Fig. 7), with ($\hbar = 1$)

$$\hat{H} = \hat{H}_{\text{TC}} + \hat{H}_{\text{vib}} + \hat{H}_{\text{H}} + \hat{H}_{\text{dis}}, \quad (21)$$

with the Tavis–Cummings, vibrational, Holstein and disorder terms

$$\hat{H}_{\text{TC}} = \sum_{n=1}^N \Delta \hat{\sigma}_n^+ \hat{\sigma}_n^- + g \sum_{n=1}^N (\hat{a}^\dagger \hat{\sigma}_n^- + \hat{\sigma}_n^+ \hat{a}), \quad (22)$$

$$\hat{H}_{\text{vib}} = \nu \sum_{n=1}^N \hat{b}_n^\dagger \hat{b}_n, \quad (23)$$

$$\hat{H}_{\text{H}} = \lambda \nu \sum_{n=1}^N (\hat{b}_n + \hat{b}_n^\dagger) \hat{\sigma}_n^+ \hat{\sigma}_n^-, \quad (24)$$

$$\hat{H}_{\text{dis}} = \sum_n \epsilon_n \hat{\sigma}_n^+ \hat{\sigma}_n^-. \quad (25)$$

Here $\hat{\sigma}_n^\pm$ are the raising/lowering operators of the n th molecule, $\Delta = \omega - \omega_C$ the detuning, λ^2 the dimensionless Huang–Rhys factor, and $g \equiv g_c/\sqrt{N}$ the single-molecule coupling with g_c collective - the same $\sqrt{N}$ structure as Eq. (11). The offsets ϵ_n have zero mean and variance W^2 .

In the single-excitation subspace and in the clean limit, $\hat{H}_{\text{TC}}$ has two polariton eigenstates $|\pm\rangle = (\hat{a}^\dagger/\sqrt{2} \pm \sum_n \hat{\sigma}_n^+/\sqrt{2N}) |0_{\text{exc+ph}}\rangle$ split by $2g_c$, plus $N - 1$ degenerate dark states of zero photon weight. Ref. [4] quantifies two effects of $\hat{H}_{\text{dis}}$: (i) **Dark states brighten:** in perturbation theory the “grey” states $|d\rangle$ acquire photon weight $\sum_d |\langle d|1_{\text{ph}}\rangle|^2 \approx W^2/g_c^2$ (plus $\approx \lambda^2 \nu^2/2g_c^2$ from the Holstein term), so strictly invisible states **appear in absorption**. (ii) **Polaritons stay protected while the coupling beats the disorder:** they are pushed

outside the disorder band by $\pm g_c$ and remain well-defined hybrid modes as long as $W \lesssim g_c$, the model crossing over to the disorder-dominated regime around $W \sim 2g_c$. The effect is *non-monotonic*: disorder-enhanced excitation transfer and vibrational entanglement *peak* at $W \sim g_c$, rather than growing without bound. Disorder is not simply detrimental.

5.3 The arrowhead-matrix picture

The reason for both effects is explained by the random **arrowhead matrix** analysis of Dubail *et al.* [3]: in the single-excitation sector the disordered Tavis–Cummings Hamiltonian is an $(N+1) \times (N+1)$ arrowhead matrix, with the random molecular energies on the diagonal and the photon - the “arrow tip” - coupled to all of them with strength $g_c/\sqrt{N}$. Two results are used here. First, the photon weight of an eigenstate ψ_a ,

$$\text{PW}_a = |\psi_{a,N+1}|^2 \equiv |\langle G, 1 | \psi_a \rangle|^2, \quad (26)$$

is $O(1)$ for the two polaritons and $O(1/N)$ for each dark state (hence $O(1)$ in total): the dark states collectively steal a finite share of the photon. Second, the dark eigenvalues *interlace* the bare molecular energies, so the polaritons are the only outliers of the band; once the disorder width exceeds the collective splitting they sink into the band and lose their photon weight. Since absorption here starts from $|A, \dots, A; 1_{\text{ph}}\rangle$, the intensity of an eigenstate *is* its photon weight: the spectra computed below are precisely the photon spectral function $A(\omega) = \sum_a \text{PW}_a \delta(\omega - \varepsilon_a)$ of Ref. [3], and the $P(v)$ heatmap is its vibronic-sector resolution.

6 Disorder in the cavity-JT model: method and implementation

6.1 How disorder enters

Following Eq. (25), disorder is added to the cavity-JT model of Sec. 4 by making each molecule’s excited-state energy random. In the JT case each molecule has *two* excited sublevels $|E_{\pm}\rangle$ which shift rigidly together, so the disorder term reads

$$\hat{H}_{\text{dis}} = \sum_n \epsilon_n \sum_{r=+,-} \hat{\sigma}_{n,r}^{\dagger} \hat{\sigma}_{n,r} = \sum_n \epsilon_n (|E_{+}\rangle \langle E_{+}|_n + |E_{-}\rangle \langle E_{-}|_n), \quad (27)$$

with $\hat{\sigma}_{n,r}^{\dagger} = |E_r\rangle \langle A|_n$ and $\hat{\sigma}_{n,r} = |A\rangle \langle E_r|_n$. Equivalently, and this is how it is implemented, the transition energy of molecule i in Eq. (9) is drawn independently from a Gaussian,

$$\epsilon_i \sim \mathcal{N}(\epsilon, W^2), \quad i = 1, \dots, N, \quad \epsilon = 7.0 \text{ eV}, \quad (28)$$

so that $\delta_i = \epsilon_i - \epsilon$ has zero mean and variance W^2 . Throughout, σ and W denote the same disorder width.

6.2 Choice of the disorder strength

Table 1 collects the energy scales. The disorder width was fixed at $W = 0.48 \text{ eV}$, i.e.

$$\frac{W}{g} \approx 2.0, \quad \frac{W}{\sqrt{N}g} \approx 1.4, \quad \frac{W}{\Omega} \approx 0.70. \quad (29)$$

In the language of Sec. 5 this places the system just *below* the $W \sim 2g_c$ crossover: nominally still strong coupling. A single strong value was chosen deliberately rather than a weak/medium/strong sweep - at weak disorder every realization simply reproduces the clean result, whereas at $W \approx 2g$ the individual realizations are physically distinct, which is what makes the case analysis of Sec. 7 informative. Figure 8 shows the distribution.

Table 1: Energy scales of the $N = 2$ cavity-JT model with disorder. All values in eV.

Quantity	Value	Quantity	Value
vibrational quantum ω	0.08196	JT coupling κ ($\kappa/\omega = 2.2$)	0.180312
cavity energy ω_c	6.85	molecular energy ϵ	7.0
Rabi splitting $\Omega = 2\sqrt{N}g$ ($\Omega/2\omega_c = 0.05$)	0.685	collective coupling $\sqrt{N}g = \Omega/2$	0.3425
per-molecule coupling $g = \Omega/2\sqrt{N}$	0.2422	Lorentzian FWHM γ	0.025
disorder width $W = \sigma$	0.48	ratios	$W/g = 1.98$, $W/\Omega = 0.70$

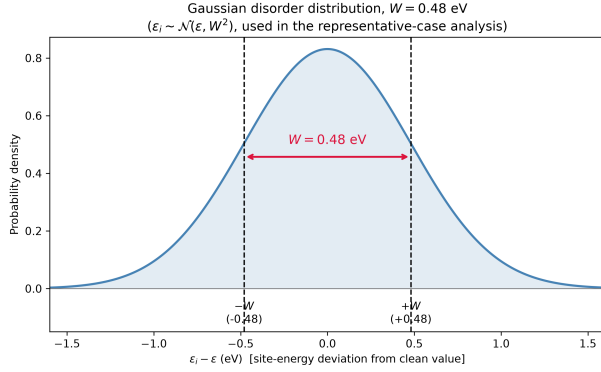


Figure 8: Gaussian disorder distribution used throughout, $\epsilon_i \sim \mathcal{N}(\epsilon, W^2)$ with $W = 0.48$ eV. Dashed lines mark $\pm W$. Absorption is averaged over 100 realizations drawn from this distribution.

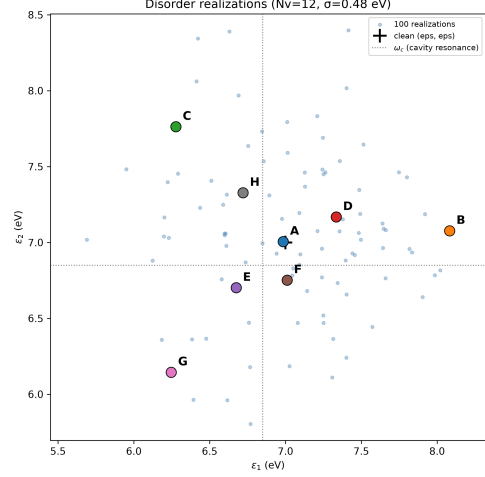


Figure 9: The (ϵ_1, ϵ_2) cloud of 100 disorder realizations at $W = 0.48$ eV. Dotted lines mark ω_c , the “+” marks the clean point (ϵ, ϵ) . Labelled points are the eight representative realizations A–H of Table 2.

6.3 Numerical procedure

For each disorder realization the following steps are carried out.

- Basis construction.** A product basis state is the 8-tuple $(e_1, n_{+1}, n_{-1}, e_2, n_{+2}, n_{-2}, p_+, p_-)$, with $e_k \in \{A, E_+, E_-\}$, vibrational occupations $0 \leq n_{\pm k} < N_v$ and photon occupations $p_{\pm} \in \{0, 1\}$. At $N_v = 12$ the full product space has 746 496 states.
- Symmetry filtering.** Only states with $n_{\text{ex}} = 1$ [Eq. (14)] and $j = -1$ [Eq. (15)], evaluated as $j = 2L_z + S_z + l_z$ with $L_z = (n_{+1} - n_{-1}) + (n_{+2} - n_{-2})$, $l_z = p_+ - p_-$ are kept, reducing the dimension to 6 900 - by a factor 108 - which is what makes a converged $N_v = 12$ calculation feasible.
- Hamiltonian assembly.** The disorder-independent part of $\hat{H}$ (vibrational, cavity, JT and matter-cavity terms) is built *once* per N_v ; only the electronic diagonal, which carries ϵ_1, ϵ_2 , is rebuilt per realization.
- Diagonalization and intensities.** $\hat{H}$ is diagonalised; the absorption intensity of eigenstate $|\psi_a\rangle$ is $I_a = |\langle i_0 | \psi_a \rangle|^2$ with $i_0 = |A, 0, 0; A, 0, 0; p_+=0, p_-=1\rangle$, i.e. both molecules in the vibronic ground state plus one RCP photon. By Eq. (26) this intensity is the photon weight of the eigenstate.
- Broadening and averaging.** Each line is convolved with a Lorentzian of FWHM γ and the result averaged over M realizations,

$$A(E) = \frac{1}{M} \sum_{m=1}^M \sum_a I_a^{(m)} \frac{1}{\pi} \frac{\gamma/2}{(E - E_a^{(m)})^2 + (\gamma/2)^2}, \quad (30)$$

on a grid of 2000 points over $E \in [6, 8]$ eV with $\gamma = 0.025$ eV and $M = 100$.

For the $P(v)$ heatmaps a second, independent route is used, following the Supplementary Material of Ref. [2]: each molecule’s JT Hamiltonian (9) is diagonalised first to obtain the vibronic eigenstates $|\lambda_i^v\rangle$ and their sector labels v ; the two-molecule plus cavity Hamiltonian is then built in that basis, restricted to $(n_{\text{ex}} = 1, j = -1)$ and diagonalised, and $P(v)$, PR follow from Eqs. (19)–(20). Since the JT eigenvectors are independent of the transition energy (disorder shifts the excited eigenvalues rigidly), the vibronic basis is computed once and reused for every realization. A state counts as **bright** when its intensity exceeds 10^{-2} of the brightest.

Because the sector-restricted dimension stays modest, everything here is obtained by *exact diagonalization*; Ref. [2] complements this with multi-configuration time-dependent Hartree (MCTDH) propagation [17] in order to reach $N = 4$ and $N = 8$, which is the route any extension of this work to larger ensembles would have to take.

6.4 Validation

Because the disorder layers were added to an existing absorption pipeline, two regression tests pin the results. The first checks that the representative-realization analysis replays exactly the same random-number stream as the disorder average - so a “case” really is one of the realizations behind that average - and that the averaged spectrum is unchanged by the addition. The second checks that the independent vibronic-basis route used for the heatmaps reproduces the eigenvalues and intensities of the direct product-basis route to $\sim 10^{-9}$, and that $\sum_v P(v) = 1$ for every eigenstate. Case A below, a near-zero-disorder draw, is a further physical control: it must - and does - reproduce the clean spectrum and heatmap.

7 Results: representative disorder realizations

7.1 The (ϵ_1, ϵ_2) cloud and the selection of cases

A disorder-averaged spectrum such as Eq. (30) hides the physics: it superposes 100 very different individual spectra. To recover the mechanism I therefore analysed the cloud of drawn pairs (ϵ_1, ϵ_2) directly (Fig. 9) and selected **eight physically meaningful realizations**, Cases A–H, each defined by a criterion on the *actually drawn* energies. Two natural coordinates organise the plane:

$$s = \frac{\delta_1 + \delta_2}{2} \quad (\text{common-mode shift}), \quad \Delta = \epsilon_1 - \epsilon_2 \quad (\text{inter-molecular mismatch}), \quad (31)$$

with $\delta_i = \epsilon_i - \epsilon$. These are orthogonal axes: s moves both molecules together relative to the photon and is compared with the collective coupling $\sqrt{N}g$, whereas Δ breaks the permutation symmetry between them and is compared with the per-molecule coupling g .

Note that W is the Gaussian’s standard deviation, not a hard bound ($\approx 32\%$ of draws fall beyond $\pm 1W$; here $38\%/35\%$ for δ_1/δ_2). Three cases (B, C, G[†]) were selected by *maximizing* a detuning or mismatch and so land, by construction, at $1.5\text{--}2.3W$ for at least one molecule - a locally stronger disorder that drives them into the localized/decoupled regime of Sec. 7, not a sampling inconsistency.

One point worth noting: a molecule is *cavity-resonant* when its brightest vibronic transition equals ω_c , which happens at $\epsilon_i \approx \epsilon = 7.0$ eV and *not* at $\epsilon_i \approx \omega_c = 6.85$ eV: ω_c was tuned to the clean bright peak, which sits ≈ 0.149 eV below ϵ because of the JT stabilisation. All criteria are therefore written in the signed detuning δ_i ; case H is a deliberate exception targeting ω_c itself. The cases are evaluated in order from a shared pool, so each claims a *distinct* realization.

Table 2 lists the cases in the order discussed below, a physical walk through the disorder plane rather than an alphabetical one: from the resonant baseline A, moving the pair together (D, E), then breaking the symmetry between the molecules by increasing amounts (F, H, C, B), and finally decoupling both from the cavity (G).

Table 2: The eight representative realizations selected from the 100-draw cloud at $W = 0.48$ eV, with their selection criteria and outcomes. $\delta_i = \epsilon_i - \epsilon$, $\Delta = \epsilon_1 - \epsilon_2$, $s = (\delta_1 + \delta_2)/2$, $g = 0.242$ eV. “#bright” is the number of polaritonic states with intensity above 10^{-2} of the maximum; the v -range lists sectors with $P(v) > 10^{-2}$ for at least one bright state. The clean $\sigma = 0$ reference is 42 bright, $\text{PR}_{\text{max}} = 19.9$, $v \in [-14, 13]$. $\dagger$: ≥ 1 molecule has $|\delta_i| > W$ (tail draw; see text).

Criterion	ϵ_1	ϵ_2	Δ	s/g	#bright	PR_{max}	v -range
A resonant baseline	6.984	7.006	0.02	0.0	39	20.6	$[-14, 13]$
D common blue detuning	7.334	7.169	0.17	+1.0	12	12.5	$[-8, 8]$
E common red detuning	6.674	6.702	0.03	-1.3	104	22.4	$[-15, 14]$
F single-mol. resonant	7.010	6.752	0.26 (g)	-0.5	132	21.6	$[-14, 13]$
H A-C line, interior	6.720	7.328	0.61 ($2.5g$)	0.0	46	16.2	$[-10, 9]$
C † opposite disorder (far)	6.277	7.763	1.49 ($6g$)	0.0	14	8.5	$[-10, 10]$
B † large mismatch (same side)	8.082	7.078	1.00 ($4g$)	+2.4	11	3.8	$[-8, 8]$
G † bare-molecule limit	6.246	6.146	0.10	-3.3	8	4.5	$[-4, 3]$

7.2 Case A - resonant baseline (effectively no disorder)

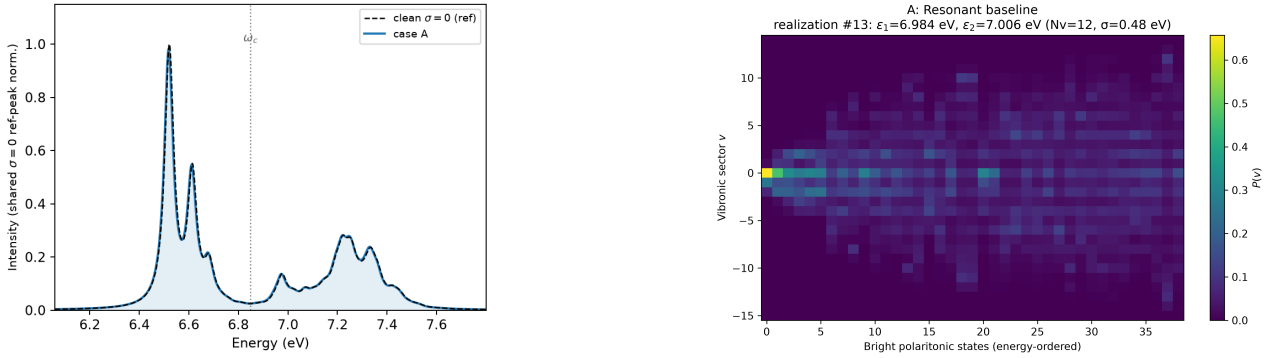


Figure 10: Case A, $(\epsilon_1, \epsilon_2) = (6.984, 7.006)$. *Left*: absorption spectrum (blue) against the clean $\sigma = 0$ reference (dashed). *Right*: $P(v)$ heatmap of the bright polaritonic states.

Both molecules are drawn essentially on resonance and matched ($\Delta = 0.02$, $s \approx 0$; Fig. 10). The spectrum reproduces the clean one peak for peak (LP at 6.518 eV with shoulders at 6.611, 6.676 eV; UP triplet at 7.222, 7.247, 7.331 eV), and the heatmap gives 39 bright states, $\text{PR}_{\text{max}} = 20.6$, $v \in [-14, 13]$, against 42, 19.9, $[-14, 13]$ for the clean reference. Case A is thus an internal control: the disorder machinery leaves the clean physics intact, and it anchors every other comparison.

7.3 Cases D and E - moving both molecules together

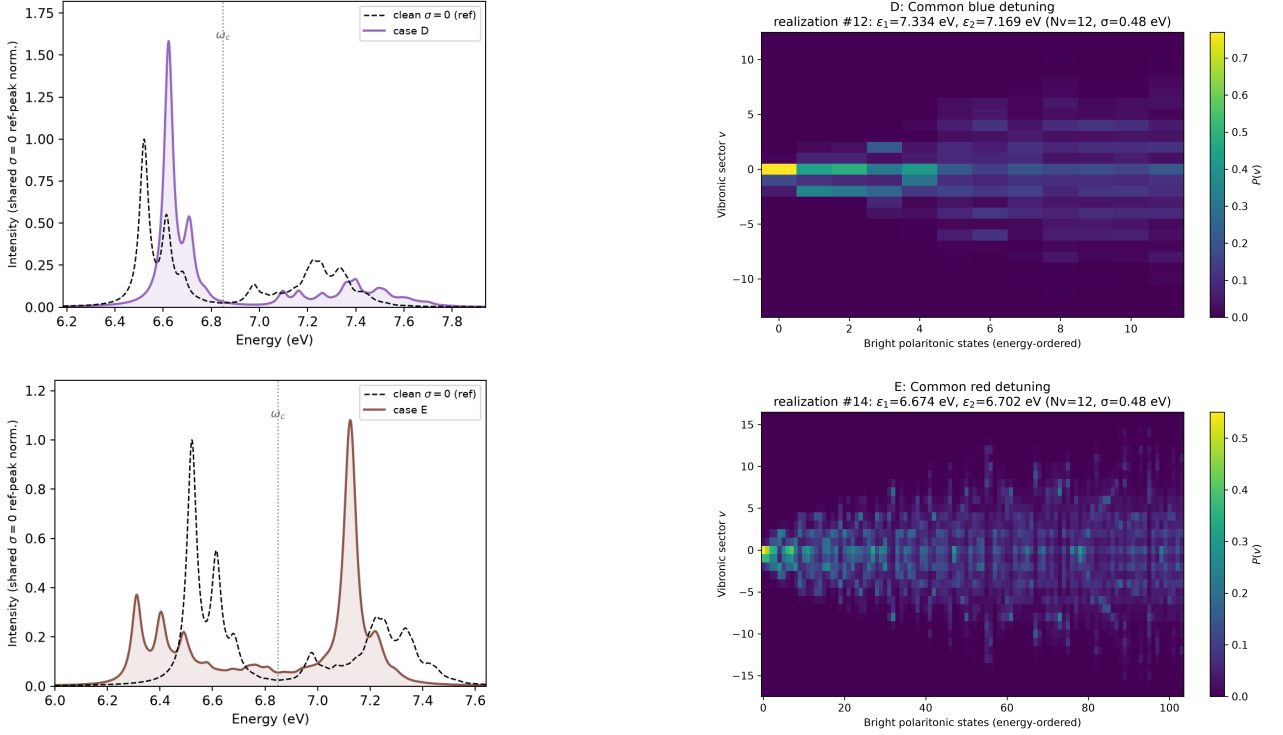


Figure 11: Top - Case D, common blue detuning, $(\epsilon_1, \epsilon_2) = (7.334, 7.169)$, $s = +1.0g$. **Bottom - Case E**, common red detuning, $(\epsilon_1, \epsilon_2) = (6.674, 6.702)$, $s = -1.3g$. Left: spectra against the clean reference (dashed); right: $P(v)$ heatmaps.

Cases D and E (Fig. 11) keep the two molecules matched (Δ small) but shift them together, blue by $s = +1.0g$ and red by $s = -1.3g$. They are nominally mirror images, but they are *not* physically equivalent.

In **Case D** the spectrum blue-shifts and collapses to two resolved LP peaks (6.623, 6.708 eV) with the UP pushed up to ≈ 7.37 eV; only 12 states stay bright, PR_{max} drops to 12.5 and the range narrows to $v \in [-8, 8]$ - the cascade is reduced to roughly 60% of baseline. In **Case E** the whole spectrum red-shifts (LP cluster at 6.311, 6.404, 6.490 eV) and the UP feature at 7.125 eV becomes the *brightest* peak, inverting the oscillator-strength ordering of A. Its heatmap is the richest of all eight cases: 104 bright states ($2.6\times$ the clean reference), $\text{PR}_{\text{max}} = 22.4$, and the widest range $v \in [-15, 14]$.

Here is a more direct way to see why. Since the molecules stay matched, the shifted pair still behaves as a single unit detuned from the cavity by s , and a detuned molecule-cavity system always splits into two branches: one that tracks the molecule's own (shifted) energy and keeps its full JT vibronic structure, and one that stays closer to ω_c and is mostly photon in character. How rich each branch looks in absorption then depends on how much of the bare JT vibronic manifold (Fig. 3, whose density is not symmetric about ϵ) sits close enough to the molecule-like branch to borrow some of its photon weight. E's red shift moves its molecule-like branch down into a denser stretch of that bare structure, so many nearby vibronic sub-levels borrow enough photon weight to cross the brightness threshold - giving the richest heatmap of the eight cases. D's blue shift moves its molecule-like branch up into a sparser stretch instead, so far fewer sub-levels light up, while the photon-like branch left behind near ω_c contributes only the two simple peaks seen in Fig. 11. Case E is the cavity-JT realisation of the disorder-enhanced regime of Ref. [4], which peaks when the offset is of order g_c .

7.4 Cases F, H, C and B - breaking the symmetry between molecules

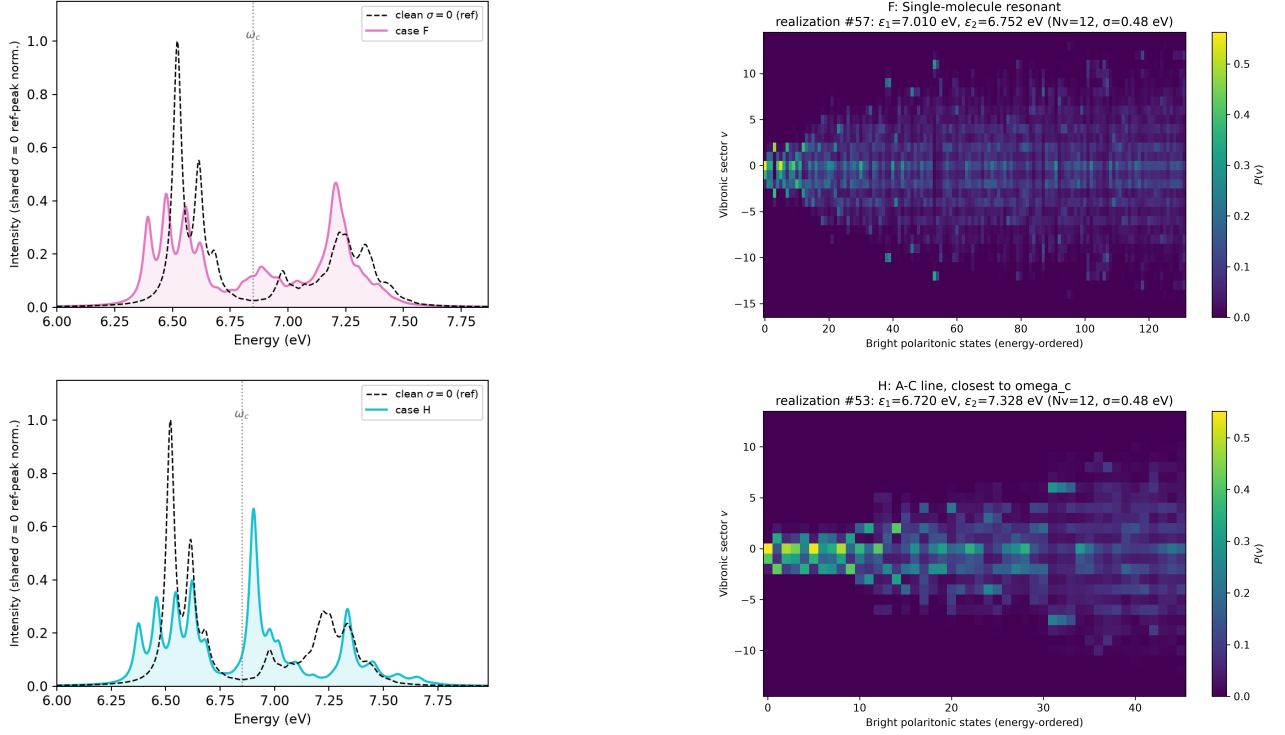


Figure 12: Top - Case F, single-molecule resonant, $(\epsilon_1, \epsilon_2) = (7.010, 6.752)$, $\Delta = g$. **Bottom - Case H**, interior point of the A-C line (opposite disorder), $(\epsilon_1, \epsilon_2) = (6.720, 7.328)$, $\Delta = 2.5g$, $s \approx 0$.

Case F (Fig. 12, top) introduces a genuine inter-molecular mismatch of exactly one coupling, $\Delta = g$: one molecule is brought back onto resonance ($\delta_1 \approx 0$) while the other stays detuned by $-g$. With $\Delta \approx g$ the system sits right at the crossover and the two molecules still hybridise collectively: 132 bright states result - the most of any case - with $\text{PR}_{\text{max}} = 21.6$ and $v \in [-14, 13]$. The spectrum is correspondingly the most structured of the set: ten resolved features spread from 6.39 to 7.39 eV, with the UP peak at 7.207 eV the brightest and the LP cluster (6.394, 6.473, 6.557, 6.619 eV) carrying 50–90% of it, so the oscillator strength is spread out rather than concentrated.

Case H was constructed on the line joining A and C in the (ϵ_1, ϵ_2) plane - the $s \approx 0$ anti-diagonal along which the two molecules are detuned *oppositely* - at the interior point closest to ω_c . Here $\Delta = 2.5g$ with $s \approx 0$. The result is mixed: 46 bright states, *more* than A's 39, so the opposite-disorder mismatch produces more peaks in absorption; but $\text{PR}_{\text{max}} = 16.2$ and $v \in [-10, 9]$, both *below* A. Disorder here simultaneously exposes more of the manifold and degrades the spread of each individual state. The brightest peak sits at 6.903 eV, essentially at ω_c , with oscillator strength spread across the whole band instead of concentrated in one polariton.

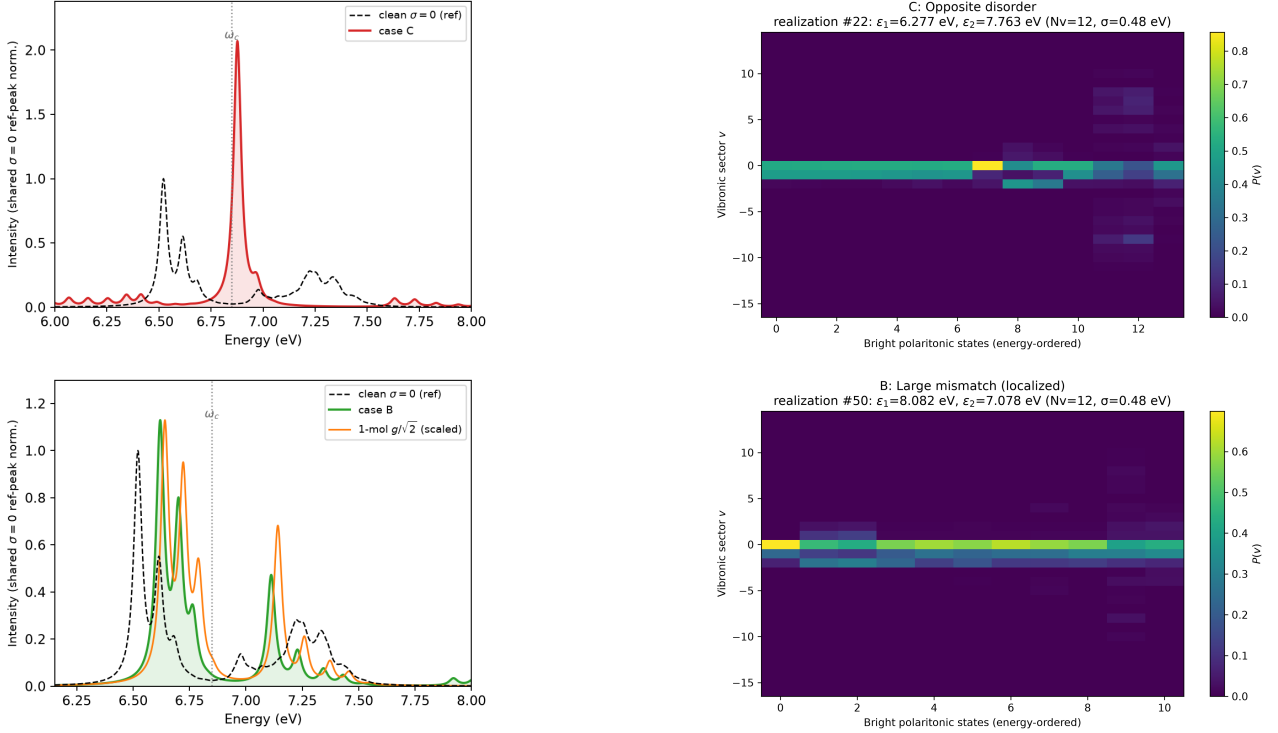


Figure 13: Top - Case C, opposite disorder pushed far out along the A–C line, $(\epsilon_1, \epsilon_2) = (6.277, 7.763)$, $\Delta = 6g$, $s \approx 0$. **Bottom - Case B**, large same-side mismatch, $(\epsilon_1, \epsilon_2) = (8.082, 7.078)$, $\Delta = 4g$, $s = +2.4g$; the orange curve is a single-molecule JT polariton spectrum computed at $g/\sqrt{2}$ for comparison.

Case C continues along the same A–H–C line to $\Delta = 6g$. Both molecules have now left the bright line, in opposite directions, and the spectrum collapses onto a single dominant peak at 6.876 eV - essentially the bare cavity photon. Only 14 states are bright, PR_{max} falls to 8.5, and the $P(v)$ weight is mostly pinned to $v \in \{0, -1, -2\}$ with a faint high- v tail. Along $A \rightarrow H \rightarrow C$ the cascade therefore weakens monotonically, $\text{PR}_{\text{max}} : 20.6 \rightarrow 16.2 \rightarrow 8.5$, as the opposite-disorder mismatch grows.

Case B takes the mismatch to the *same* side instead: molecule 1 is detuned by $+4.5g$ while molecule 2 stays near resonance ($\Delta = 4g \gg g$), and the excitation *localizes* entirely on the resonant molecule. The spectrum is that of a *single* JT polariton - LP at 6.620 eV followed by a clean vibrational progression at 6.701, 6.763 eV and one UP feature at 7.113 eV - and it overlays the independently computed one-molecule spectrum at $g/\sqrt{2}$ (orange curve, Fig. 13). The heatmap confirms this: 11 bright states, all with $P(v)$ confined to a flat horizontal band, $\text{PR}_{\text{max}} = 3.8$, i.e. the single-molecule bound of Sec. 3. Case B is thus a direct, disorder-induced crossover from the collective physics of Ref. [2] back to the single-molecule physics of Ref. [1]; its 11 bright lines are a Franck–Condon progression, not a cascade.

7.5 Case G - bare-molecule limit

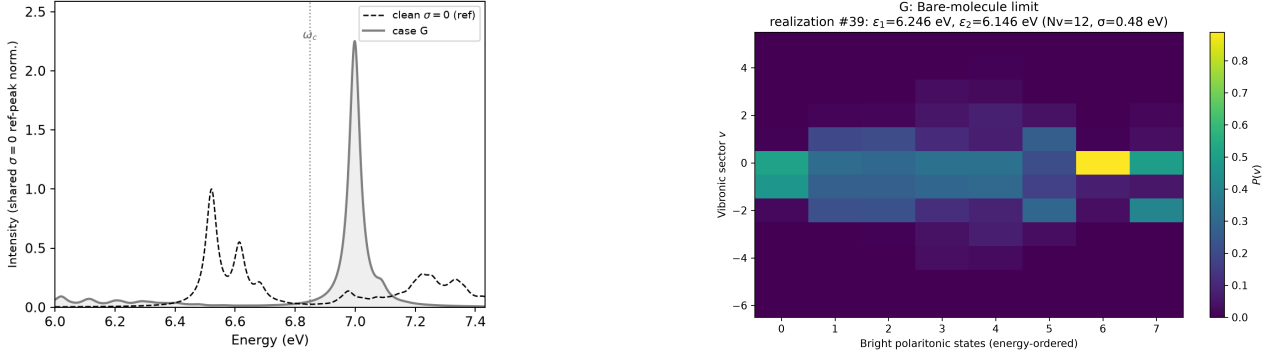


Figure 14: Case G, bare-molecule limit, $(\epsilon_1, \epsilon_2) = (6.246, 6.146)$: both molecules pushed far below resonance together, $s = -3.3g$, $\Delta = 0.10$.

Finally, Case G (Fig. 14) pushes *both* molecules far below resonance together: they remain perfectly matched ($\Delta = 0.10$) but sit $s = -3.3g$ - well outside the polariton band - from the bright line. The absorption collapses to a single peak at 6.998 eV, the essentially uncoupled cavity photon, and the heatmap retains only 8 bright states confined to $v \in [-4, 3]$, one of which holds $\approx 90\%$ of its weight in $v = 0$; $\text{PR}_{\text{max}} = 4.5$.

Cases B and G are both “dead” at nearly the same $\text{PR}_{\text{max}} \approx 4$, but they die by *different* mechanisms - a distinction that is one of the main results of this work. B dies along the *mismatch* axis: $\Delta \gg g$ decouples the molecules from each other and the excitation localizes on one, giving a bounded single-molecule cascade. G dies along the *common-mode* axis: the molecules stay perfectly matched, but $|s| \gg \sqrt{N}g$ places both outside the polariton band, so the bright eigenstate reverts to the bare photon and the cavity-mediated exchange driving the cascade is switched off.

7.6 Overview: bright states and cascade strength across cases

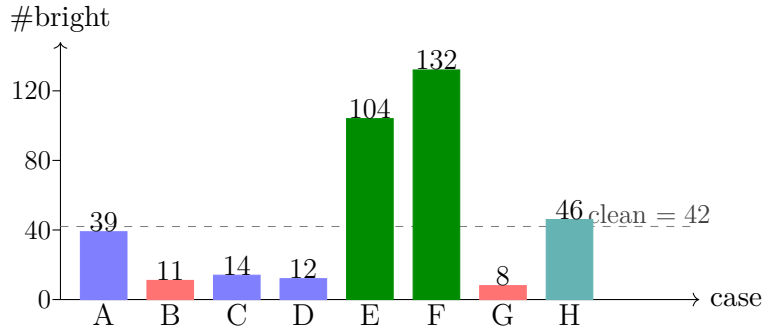


Figure 15: Number of bright polaritonic states per case (intensity above 10^{-2} of the maximum), compared with the clean $N = 2$ reference (dashed, 42). Green: enhanced (E, F); red: dead (B, G); teal: intermediate (H); blue: the remaining cases (A, C, D).

Figure 15 summarises the bright-state count. Ordering the cases instead by the cleaner, per-state cascade metric PR_{max} gives

$$E (22.4) \approx F (21.6) \approx A (20.6) > H (16.2) > D (12.5) > C (8.5) > G (4.5) \approx B (3.8).$$

This ordering also separates cleanly into three groups rather than forming a smooth trend: E, F and A cluster at or above the clean value; H, D and C form a middle band; and G, B sit together at the bottom, an order of magnitude lower. Neither grouping lines up with the mismatch Δ alone: Case G’s mismatch ($\Delta = 0.10$) is far smaller than Case H’s ($\Delta = 2.5g$), yet G is dead and H is only partially degraded. This three-way split is itself a hint that a single disorder magnitude does not set the outcome; Sec. 8 makes precise what does.

8 Discussion

Every case above can be described through a single candidate mechanism, **bright/dark mixing**, adapted from the random-arrowhead-matrix analysis of Ref. [3]. That analysis was carried out for the disordered Tavis–Cummings (Holstein-type) model, where each molecule contributes a single excited level; our JT molecules instead carry a full vibronic cascade, and extending the arrowhead proof to that richer setting has not been done here. With that caveat, the qualitative picture is as follows. In the clean, matched, resonant system the permutation symmetry splits the two-molecule matter states into a bright symmetric combination $(|E, A\rangle + |A, E\rangle)/\sqrt{2}$, which carries all the photon coupling and forms the LP/UP pair split off by $\pm\sqrt{N}g$, and a dark antisymmetric combination with zero photon weight and hence zero intensity - this part is exact, independent of the vibronic richness (Sec. 1).

Whether this exact, clean-limit coupling is actually what drives the case-by-case behaviour of Sec. 7, rather than merely being consistent with it, is left open (see Caveats below). The key point from that case-by-case evidence is that the outcome is *not* monotonic in the disorder magnitude. Moderate mixing - the bright polariton pushed *towards* the dense dark manifold, as in E and F - is a plausible reason why more of the cascade becomes visible than in the clean system, matching the enhancement Ref. [4] describes for the simpler Holstein-type model. Excessive mixing, or mixing in the wrong direction, coincides with the cascade being destroyed instead. Which of the two happens is governed by proximity to cavity resonance and to the collective coupling scale, not by $|\delta_1|, |\delta_2|$ alone.

Caveats. Three limitations should be kept in mind. (i) Each case is a *single* extremal draw chosen to illustrate a mechanism; PR and #bright fluctuate strongly between realizations, so quantitative claims would require averaging over all realizations at fixed W . (ii) #bright depends on the brightness threshold. (iii) The “bright/dark” language is borrowed from the $N \rightarrow \infty$ arrowhead-matrix limit of Ref. [3], itself proved for the disordered Tavis–Cummings (Holstein-type) model rather than a JT vibronic cascade; extending that proof to our model, and to the $N = 2$ case specifically, remains to be done. With two molecules there is only one dark combination per vibronic channel, so “brightening” means antisymmetric channels gaining photon weight, not the filling of a large dark band.

9 Summary and conclusions

1. **Reproduced** the single-molecule cavity Jahn–Teller polariton spectrum of Ref. [1] - LP/UP branches, bounded vibronic sectors $v \in \{0, -1, -2\}$ - and the two-molecule collective spectrum of Ref. [2] together with its $P(v)$ heatmap, recovering the unbounded cascade with PR up to ≈ 20 over 42 bright states.
2. **Introduced static Gaussian energy disorder** into the $N = 2$ cavity-JT model, following the disordered Holstein–Tavis–Cummings [4] and random-arrowhead [3] frameworks: $\epsilon_i \sim \mathcal{N}(\epsilon, W^2)$, with the spectrum built from the photon weight of each eigenstate of the $(n_{\text{ex}} = 1, j = -1)$ block, Lorentzian broadened and averaged over realizations.
3. **Implemented and analysed** the disorder layer: symmetry filtering from 746 496 to 6 900 basis states, a representative-realization analysis recovering the individual draws behind the average, eight physically selected cases and a $P(v)$ heatmap for each, validated by regression tests and by the near-zero-disorder control (Case A).
4. **Key result.** Whether disorder destroys the collective vibronic cascade depends on whether it pushes the bright polariton *closer to* the dense manifold of otherwise-dark states or *pulls it away* - governed by proximity to cavity resonance and to the collective coupling scale, not by the disorder magnitude alone. In the favourable direction disorder

enhances the observable cascade (Cases E and F: 104 and 132 bright states versus 42 for the clean system, at undiminished PR); in the unfavourable ones it destroys it by two distinct routes, localization on a single molecule (Case B) and decoupling from the cavity (Case G).

Natural continuations are to map PR over the mismatch and common-mode shift more broadly at several disorder widths, to separate convergence in the vibrational cutoff N_v from the disorder effect, and to extend the analysis to $N > 2$, where the dark manifold becomes a genuine band and the arrowhead predictions apply quantitatively.

References

- [1] K. R. Nandipati and O. Vendrell, *Cavity Jahn-Teller polaritons in molecules*, Phys. Rev. A **107**, L061101 (2023).
- [2] S. K. Pandit, A. Pandey, A. Shankar and K. R. Nandipati, *Collective Vibronic Cascade in Cavity-Coupled Jahn-Teller Active Molecules*, arXiv:2511.07880 [quant-ph] (2025).
- [3] C. Dubail, T. Botzung, J. Schachenmayer, G. Pupillo and D. Hagenmüller, *Large random arrowhead matrices: multifractality, semi-localization, and protected transport in disordered quantum spins coupled to a cavity*, arXiv:2105.08444 (2021); Phys. Rev. A **105**, 023714 (2022).
- [4] D. Wellnitz, G. Pupillo and J. Schachenmayer, *Disorder enhanced vibrational entanglement and dynamics in polaritonic chemistry*, Commun. Phys. **5**, 120 (2022).
- [5] H. A. Jahn and E. Teller, *Stability of polyatomic molecules in degenerate electronic states. I. Orbital degeneracy*, Proc. R. Soc. Lond. A **161**, 220 (1937).
- [6] H. C. Longuet-Higgins, U. Öpik, M. H. L. Pryce and R. A. Sack, *Studies of the Jahn-Teller effect. II. The dynamical problem*, Proc. R. Soc. Lond. A **244**, 1 (1958).
- [7] I. B. Bersuker, *The Jahn-Teller Effect* (Cambridge University Press, Cambridge, 2006).
- [8] H. Köppel, W. Domcke and L. S. Cederbaum, *Multimode molecular dynamics beyond the Born-Oppenheimer approximation*, Adv. Chem. Phys. **57**, 59 (1984).
- [9] R. L. Whetten, K. S. Haber and E. R. Grant, *The dynamic Jahn-Teller effect in sym-triazine: Nonadiabatic wave functions and hindered fluxionality*, J. Chem. Phys. **84**, 1270 (1986).
- [10] D. C. Harris and M. D. Bertolucci, *Symmetry and Spectroscopy: An Introduction to Vibrational and Electronic Spectroscopy* (Dover, New York, 1989).
- [11] T. W. Ebbesen, *Hybrid light-matter states in a molecular and material science perspective*, Acc. Chem. Res. **49**, 2403 (2016).
- [12] J. Feist, J. Galego and F. J. Garcia-Vidal, *Polaritonic chemistry with organic molecules*, ACS Photonics **5**, 205 (2018).
- [13] F. Herrera and F. C. Spano, *Cavity-controlled chemistry in molecular ensembles*, Phys. Rev. Lett. **116**, 238301 (2016).
- [14] S. M. Barnett, R. P. Cameron and A. M. Yao, *Duplex symmetry and its relation to the conservation of optical helicity*, Phys. Rev. A **86**, 013845 (2012).
- [15] M. D. Crisp, *Jaynes-Cummings model without the rotating-wave approximation*, Phys. Rev. A **43**, 2430 (1991).
- [16] K. R. Nandipati and O. Vendrell, *Dynamical Jahn-Teller effects on the generation of electronic ring currents by circularly polarized light*, Phys. Rev. Research **3**, L042003 (2021).
- [17] O. Vendrell, *Coherent dynamics in cavity femtochemistry: Application of the multi-configuration time-dependent Hartree method*, Chem. Phys. **509**, 55 (2018).
- [18] C. R. Harris *et al.*, *Array programming with NumPy*, Nature **585**, 357 (2020).
- [19] P. Virtanen *et al.*, *SciPy 1.0: fundamental algorithms for scientific computing in Python*, Nat. Methods **17**, 261 (2020).
- [20] J. D. Hunter, *Matplotlib: A 2D graphics environment*, Comput. Sci. Eng. **9**, 90 (2007).